ADVANCED DATABASE SYSTEMS PORTFOLIO

Project Title: Design and Implementation of a Library Management Database System

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Course: Advanced Database Systems
Selected Organisation: Library Management System

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1. Introduction

This project involved the design, implementation, testing and evaluation of an advanced database system based on a Library Management System. The purpose was to apply database concepts and practical techniques to develop a system capable of managing library operations, including borrowers, books, staff, loans and fine payments.

The project used MySQL for relational database implementation and MongoDB for NoSQL database implementation. MongoDB Atlas was also used to demonstrate cloud database deployment and distributed database concepts.

The project was divided into five portfolios, each focusing on a different area of advanced database management.

2. Portfolio Activities and Implementation

Portfolio 1: Database Design and Modelling

The first portfolio focused on designing a relational database for the Library Management System using MySQL.

The database consisted of six main tables: Borrower, Book, Staff, Loan, FinePayment and FineRate. Primary keys and foreign keys were implemented to establish relationships between the tables and maintain referential integrity.

Constraints such as NOT NULL, UNIQUE, CHECK and DEFAULT were applied to improve data accuracy and reliability. An Entity Relationship Diagram (ERD) was also created to illustrate the database structure and relationships.

The database design considered normalisation principles to reduce unnecessary data duplication and improve consistency.

Portfolio 2: Query Processing and Optimisation

The second portfolio focused on writing and evaluating SQL queries.

Queries were developed to retrieve current loans, overdue loans and outstanding fines. JOIN operations, aggregate functions and filtering conditions were used to retrieve relevant information from multiple tables.

The MySQL EXPLAIN command was used to examine query execution plans. An index was created on the Loan table using the ReturnDate and DueDate columns to investigate query optimisation.

The before-and-after execution plans were captured and analysed to understand how indexing can influence query performance. The exercise also demonstrated that the database optimiser considers table size and query cost when selecting an execution plan.

Portfolio 3: Transactions and Concurrency

The third portfolio explored transaction management and concurrency control in MySQL.

Transaction examples were implemented to demonstrate COMMIT and ROLLBACK operations. These exercises showed how database changes can either be permanently saved or cancelled when necessary.

Isolation levels were examined to understand how transaction settings affect the visibility of data between concurrent operations. Concurrency examples and control mechanisms were also explored to demonstrate how database systems manage simultaneous access and protect data integrity.

Screenshots and execution outputs were captured as evidence of the transaction tests.

Portfolio 4: NoSQL and Advanced Data Models

The fourth portfolio involved implementing a NoSQL database using MongoDB.

A MongoDB database named LibraryManagement was created with two collections: Borrowers and Books.

Borrower documents included embedded loan information, demonstrating how related data can be stored within a single document. Book information was stored separately in the Books collection.

Several MongoDB queries were executed using MongoDB Compass. These included retrieving all borrowers, searching for a specific borrower, identifying overdue borrowers, finding current loans and retrieving available books.

The exercise demonstrated the flexibility of document-oriented databases and the differences between relational and NoSQL data models.

Portfolio 5: Distributed and Cloud Database Exercise

The fifth portfolio focused on cloud database deployment and distributed database concepts using MongoDB Atlas.

A cloud database named LibraryManagementCloud was created with Borrowers and Books collections. Sample documents were added to demonstrate cloud-hosted data storage and access through MongoDB Compass.

The db.hello() command was executed to inspect the Atlas replica-set configuration. The output identified three replica-set hosts and showed the current primary member through the isWritablePrimary field.

A consistency test was also performed using majority write concern. A test document was inserted and successfully retrieved from the cloud database.

Sharding was explored through a proposed architecture diagram showing how data could be distributed across multiple shards. Sharding was not deployed in the exercise; the diagram represented a proposed design for a larger database environment.

3. Skills and Knowledge Gained

The project improved my practical understanding of database design, implementation and administration.

Some of the major skills developed include:

- Designing relational databases and identifying relationships between entities.
- Applying primary keys, foreign keys and database constraints.
- Writing SQL queries involving joins, filtering and aggregation.
- Analysing query execution plans and understanding indexing.
- Managing database transactions and concurrency.
- Creating and querying MongoDB document collections.
- Understanding embedded documents and NoSQL data modelling.
- Deploying a database to MongoDB Atlas.
- Understanding replica sets, majority write concern and cloud database architecture.
- Organising technical evidence and maintaining project documentation using GitHub.

4. Individual Reflection

Working on this project provided practical experience in applying database concepts to a realistic library management scenario.

One of the most valuable aspects was learning how to design a relational database and enforce data integrity using keys and constraints. Testing the SQL queries also improved my understanding of how databases retrieve and process information.

The MongoDB exercises helped me understand that not every application requires a traditional relational database. Document-oriented databases provide flexibility when dealing with records containing nested or varying information.

The cloud database exercise also introduced me to database deployment beyond a local computer. I gained an understanding of replica sets, primary nodes, majority write concern and the purpose of sharding in larger database environments.

Some challenges were encountered during implementation, particularly when configuring database connections, testing constraints and understanding transaction behaviour. Working through these challenges improved my troubleshooting and problem-solving skills.

Overall, the project strengthened my confidence in using MySQL, MongoDB Compass, MongoDB Atlas and GitHub to develop, test and document database solutions.

5. Conclusion

The Advanced Database Systems portfolio successfully demonstrated the application of database concepts through five practical exercises.

The project covered relational database design, SQL query processing and optimisation, transaction management, NoSQL data modelling, and distributed and cloud database concepts.

The use of MySQL and MongoDB provided experience with both relational and document-oriented database technologies. The MongoDB Atlas exercise further demonstrated how cloud database services can support scalable and highly available applications.

The project provided a strong practical foundation in advanced database development and management.

6. References

1. Oracle. *MySQL Reference Manual*. https://dev.mysql.com/doc/
2. MongoDB. *MongoDB Manual*. https://www.mongodb.com/docs/manual/
3. MongoDB. *MongoDB Atlas Documentation*. https://www.mongodb.com/docs/atlas/
4. Elmasri, R. and Navathe, S. B. *Fundamentals of Database Systems*. Pearson.

7. AI Use Declaration

AI assistance was used during this project to support learning, clarify database concepts, troubleshoot implementation issues, improve explanations and organise documentation.

The database implementation, query execution, testing and evidence collection were carried out as part of the practical project work. The submitted results and screenshots represent the work performed during implementation.

All AI-assisted content was reviewed and should be checked for accuracy before submission.